Socratic-SWE: Self-Evolving Coding Agents via Trace-Derived Agent Skills

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HarnessRSI 64 ★★

skills + weight training

Paper v1

1 What changes, what is kept

Solver traces
Inspection, edits, commands, tests;
keep both successful and failed traces.

▼

Distill structured skills
Qwen3.6-27B → name, description,
conditions, ordered operations; dedup/filter.

▼

Generate repository repair tasks
Skill + repository + Solver evidence;
reversible bug, original code = reference fix.

▼

Hard gate: all checks pass
Format → grounding → execution stability
→ semantics / at least one valid repair.

▼

Reward the Generator
Valid × cosine(task gradient, reference);
100 BeyondSWE tasks define the direction.

▼

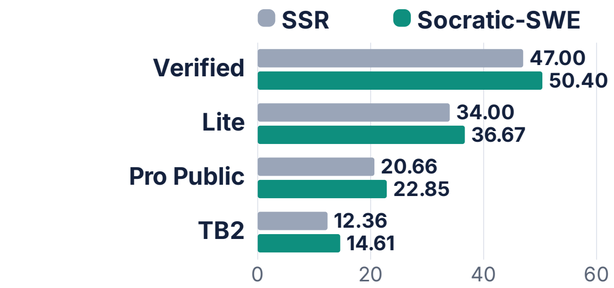
Train the shared policy
Generator: GRPO · Solver: GDPO;
full pass / partial repair / no regressions.

Updated weights → new traces → next curriculum
Not a weights-frozen skill-memory experiment

Acceptance ≠ alignment cutoff: Eq. 2 / Algorithm 1 admit Valid = 1 tasks; alignment is a training reward. No extra cutoff is specified.

Unresolved: seed-task trace collection vs repository-only initialization; Algorithm 1 supplies a registry but omits its refresh.

2 Final round leads on every benchmark



Pass rates (%) · Table 1 · Qwen3.5-9B · 36k instances. SSR ties Socratic-Zero on Terminal-Bench.

Verified: base 42.60 → 50.40; +7.80 vs base, +3.40 vs SSR. No seed/run uncertainty reported.

4 Where gains are small / unproven

- **Gains span benchmarks:** +7.80 / +7.00 / +5.61 / +4.50 over base; larger on Verified/Lite. No per-repo breakdown; one Solver model.
- **Early tie:** round-1 terminal 12.36 equals later SSR and Socratic-Zero. Overall 31.13 is a benchmark mean, not a seed mean.
- **Small knobs:** hybrid 50.60 vs 50.40; extractors 49.80 / 50.40 / 51.00. No variance/CI.
- **Plateau:** five-round Verified 52.00 vs SSR 48.00; fixed repos, beyond the 36k headline budget.
- **Leakage:** training/evaluation repo overlap unknown; no benchmark contamination audit. Oracle-text safeguards are narrower.

3 Controls and comparison limits

Verified · round 3 (Table 2)	%
Full system	50.40
No Skill Registry	46.20
Manual skills, no trace distillation	48.00
GRPO instead of GDPO	48.60

Reward ablation (Tables 3 / 6): hardness 47.40; uncertainty 48.20; variance 48.80; inner product 49.40; cosine 50.40. Hybrid 50.60 exceeds default.

Baseline access: Absolute-Zero has execution verification; SSR has execution consistency + binary rewards. Socratic-Zero has a 397B Teacher and task-design distillation. No same trace-registry grant reported.

SPIRAL: defect/repair self-play + role advantages. R-Zero: Challenger tasks + majority vote. Seed-requiring baselines get 10% SWE-smith; allocation per method unspecified.

Budget: matched instances/interactions stated; comparable SSR FLOPs claimed. No baseline tuning budgets or itemized extractor/Teacher cost.

Default selection: BeyondSWE 100-task gradients shape curriculum, not Solver training. Size sensitivity uses Verified (49.20 / 50.40 / 50.60 at 20 / 100 / 200 tasks); no separate tuning split stated.

Cost (Table 5): 15.0 h/round on 8×A100; 1.3 h alignment overhead. Three-round budget 45 h. Distillation cost not separately listed.

Reporting limit: no seed/run counts or variance for main results or ablations. Gradient-stability reruns are not benchmark repeats.

Takeaway

Trace-derived skills improve a repair-task curriculum under stated matched training budgets. The registry ablation supports its role — but shared weights also change, so this is not evidence for a frozen-model harness gain alone.

Our read · HarnessRSI

64 · ★★ · recorded level: retained
Reports: transfer, ablations, runtime.
Missing: variance, overlap audit, tuning budgets.
Appendix F adds cost to the stored judgment.